



Ride-Share Database: Full-Stack Web Application

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Computer Science 3380:

Database Applications and Information Systems

https://github.com/ssyht/Rideshare_DB-Sanjit--ssyht-

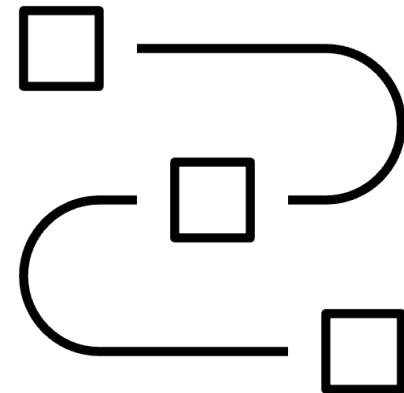


Motivation

Transportation systems today lack simple, fast, and reliable ways to book shared rides for small groups. Students, daily commuters, and community members often rely on informal coordination (texts, group chats), which leads to:

- **Uncertainty about availability**
- **No centralized ride listings**
- **No reservation tracking**
- **No structured payment workflow**

Build a lightweight yet functional ride-sharing platform where users can browse trips, reserve seats, make payments, leave feedback, and manage reservations.



Problem Statement

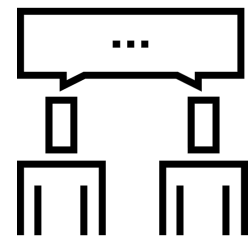
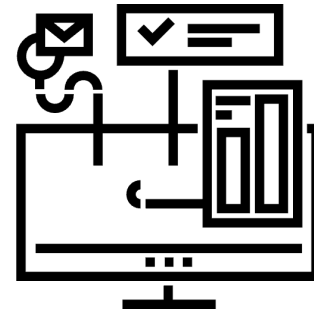
Users need a system that provides:

- A simple way to **search available trips**.
- A secure method to **reserve seats**.
- An integrated **payment workflow**.
- The ability to **cancel or modify bookings**.
- A structured way to **leave feedback**.

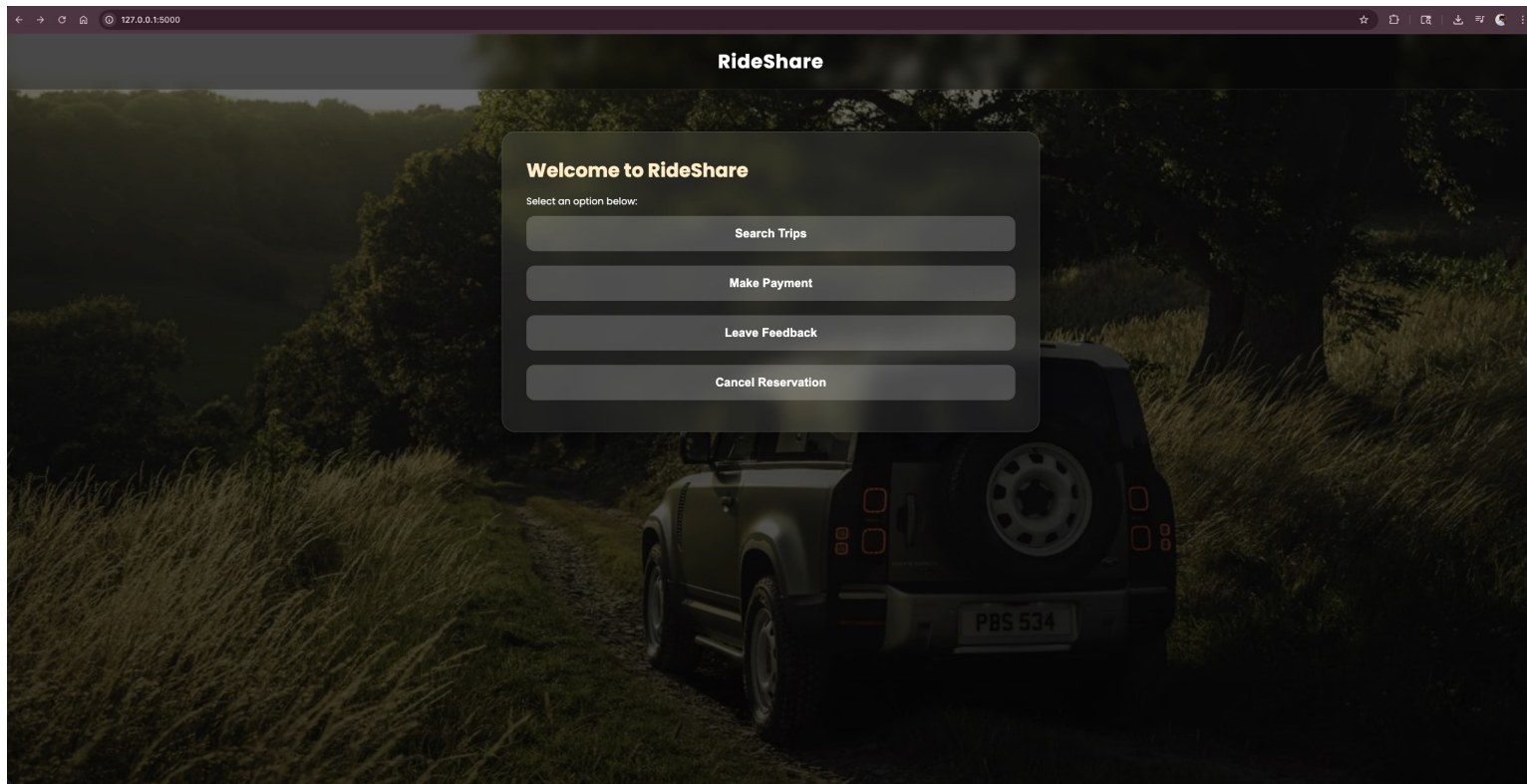
But existing small-scale ride-sharing systems are either too complex or non-existent.

Statement:

"How can we design a minimal, intuitive, and fully functional ride reservation system with database support?"

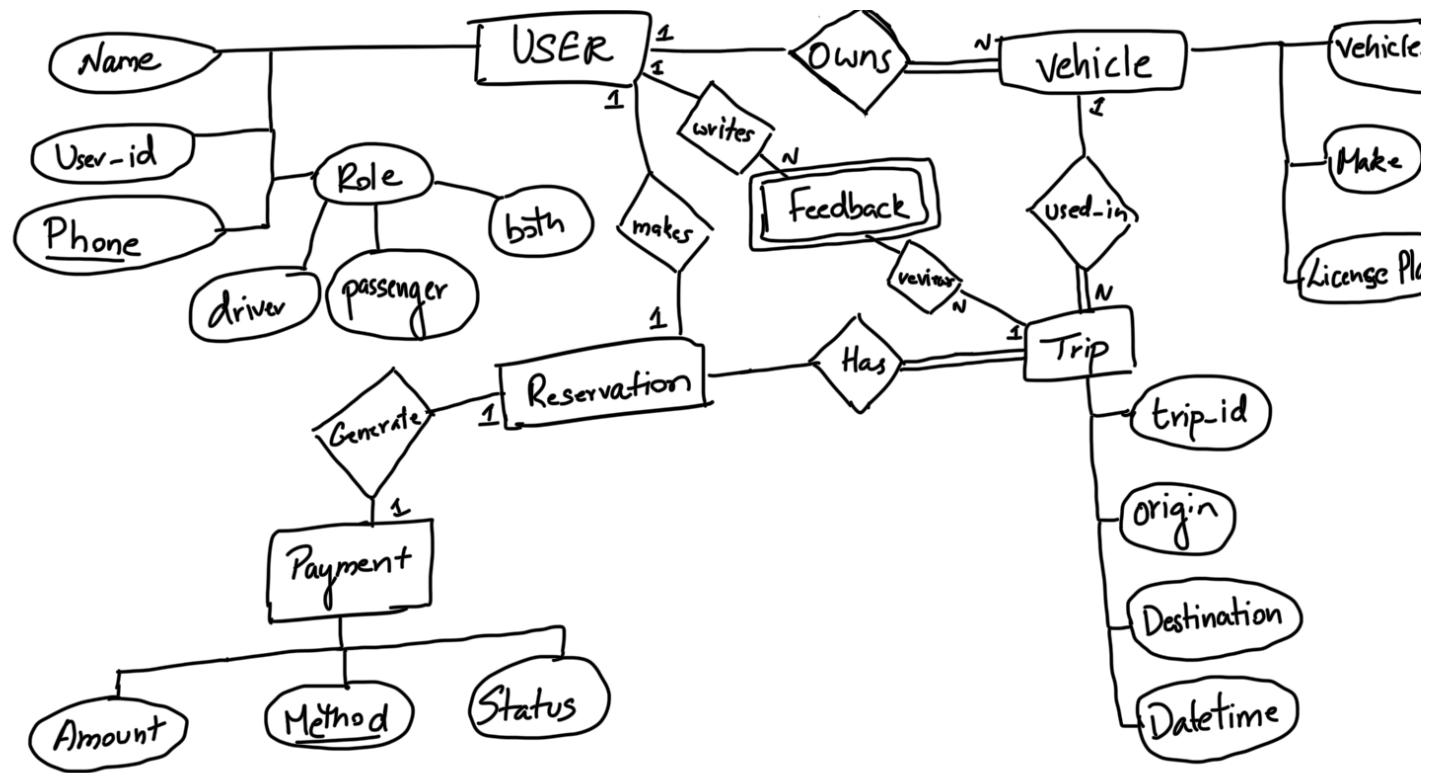


Demo



<https://github.com/ssyht/Rideshare> DB-Sanjit--ssyht-

ER Model



System Architecture

Frontend (UI Layer):

- HTML/CSS
- Forms for user input

Backend (Application Layer):

- Flask routes for each user interaction
- Input validation
- Business logic for reservation, payments, and cancellations.

Database Layer (MySQL)

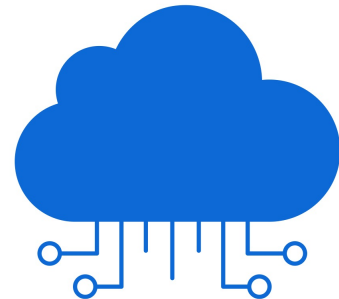
- Trips
- Reservations
- Payments
- Users
- Feedback



Data flows from UI → Flask → MySQL → Flask → UI

Future Enhancements

- User authentication & login
- Real-time seat availability
- Email/SMS confirmation
- Admin dashboard for managing trips
- Mobile app version
- Dynamic pricing model
- Map integration for origin/destination



Conclusion

This project demonstrates the successful integration of:

- Backend API logic (Flask)
- Database design (MySQL)
- Modern frontend UI
- Core ride-sharing functionalities

It is a compact but complete demonstration of a production-style system, built from scratch and fully testable.